Heterogeneous $J_c(T,H)$ literature

n = 934 articles retrieved
and screened, Elsevier + Springer;
fitted curves also come from arXiv
preprints and NHMFL records

Applicability window, Eq. (1)

Partial-fit and full-fit

Cross-model agreement

Cohort A: $J_c(T)$ at fixed H
Cohort B: $J_c(H)$ at fixed T

$T/T_c < 0.7$ $\Delta H/H_{c2} > 0.3$

50 papers contributing fitted curves
35 compound labels, 27 composition keys
3303 extracted critical-current points
20 compounds fittable on both axes
257 / 52 β_T / β_H fits, the latter from 12 papers
96 per-paper anchors over 32 papers

K ≥ 3 anchors (per regime)

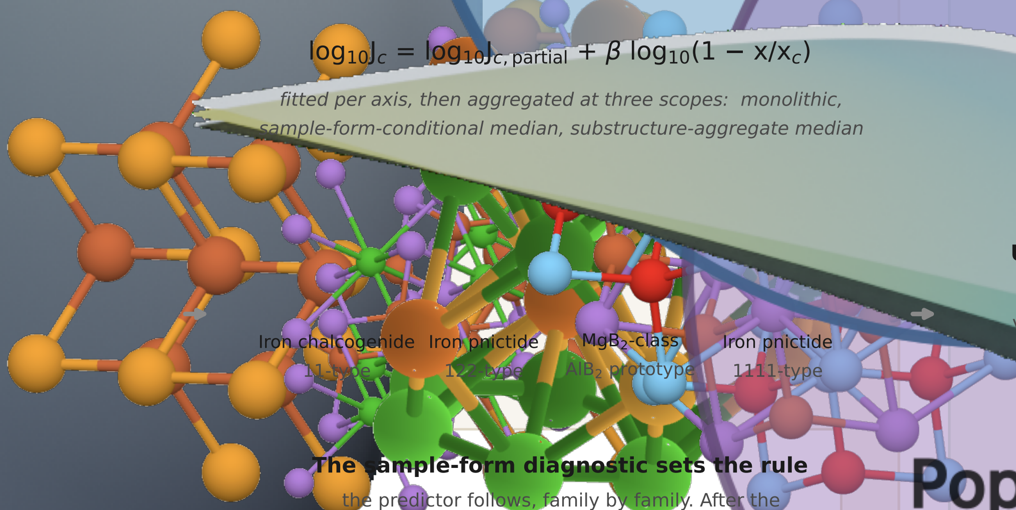
Multi-stage substructure conditional aggregation

the family label accounts for 0.52 of the temperature-exponent variance, p = 0.007

source papers are the permutation unit. On the field axis it is 0.038 at p = 0.98 and no family-level verdict is given

$$\log_{10} J_c = \log_{10} J_{c,\,\mathrm{partial}} + \beta \log_{10}(1 - x/x_c)$$

fitted per axis, then aggregated at three scopes: monolithic, sample-form-conditional median, substructure-aggregate median



Iron chalcogenide 11-type | Iron pnictide 122-type | MgB_2-class AlB_2 prototype | Iron pnictide 1111-type

The sample-form diagnostic sets the rule

the predictor follows, family by family. After the repair of Sec. III.F it explains 0.81, 0.37 and 0.04 of the within-family scatter in the anchor, for the three families above in that order.

It does not establish the distinction to significance: sample form is nearly a relabelling of source paper here, and under the only null that respects that structure the MgB_2 cell can never return p below 0.33.

No fold improvement is reported either. Under leave-one-substructure-out the errors are 12.3 without conditioning and 14.9 with it over seven families, 1.19 and 0.55 on the fits passing physicality over three.

Bounded screening grade predictions

Screening-grade predictions

Universal $(T/T_c,\ H/H_{c2})$ scaling is not adopted

within-bin standard deviation falls 13.0%, 24.3% on the variance scale, both below the 30% adoption threshold, on the pre-withdrawal 23-compound cohort (Table III)

Populated su… (three families qu…

Family-scope $J_c(T,H)$ envelope with a 95% bootstrap interval

183 candidate compounds evaluated
84 dispatched, every one MgB_2-class
60 refused a field target, no $H_{c2,0}$ anchor
4.98 $\log_{10} J_c$ at 4.2 K and 5 T

One family curve per grid point; no per-compound ranking.